Group Assignment Proposal

1. Group Information

Item	Details
Intake Code	APD2F2602CS(DF), APD2F2602CS(CYB), APU2F2602CS(DF)
Group Number	GROUP 7

No.	Student ID	Student Name	Main Role in Project
1	TP076493	BHAVANPREET KAUR SOHI A/P JASBIR SINGH	Member
2	TP076766	CHANG JIR SHEN	Member
3	TP075848	GAN ZHI MING	Member
4	TP076502	MISHA A/P PATHMARAJAH	Member
5	TP077059	RUVIEKA A/P M BALAKRISHNAN	Leader

2. Website / Web Application Title

Item	Description
Proposed Website Title	Muzik 67
Brief Description of the Website	Interactive music-learning web application designed for children aged 6–7. It introduces foundational music concepts through colourful lessons, listening activities, educational games, quizzes and a virtual keyboard. Children can practise recognising notes, rhythms and instrument sounds while earning scores and tracking their learning progress.
Main Learning Area / Subject Covered	Foundational music education, including pitch and note recognition, rhythm and beat counting, basic music notation, instrument

	recognition, ear training, virtual keyboard practice and simple melody creation.

3. Objectives

List the main objectives of the proposed web application (maximum 3 objectives only).

No.	Objective
1	To use interactive listening and tapping activities to teach kids the fundamentals of melody and rhythm.
2	To use guided, gamified practice to assist kids learn a new musical ability (identifying notes and performing simple tunes).
3	To introduce kids to a musical instrument by letting them explore and perform it within the app, such as a virtual keyboard.

4. Mission Statement

Item	Description
Mission Statement	Provide an engaging and interactive online learning platform that introduces children aged 6–7 to the fundamentals of music in a fun, simple, and age-appropriate way.
Problem / Need Addressed by the Website	1. Lack of engaging music-learning resources for young children 2. Difficulty maintaining children's attention 3. Music concepts can be difficult for beginners to understand
How the Website Supports Learning	Interactive games, MCQs, image drag and drop

5. Targeted Users / Audience Modelling

User Type	**Description of User**	**User Needs / Expectations**	**Access Level** **(public access/ login required)**
Admin	The manager/owner who has complete control over the platform's users and content	Lessons and songs must be managed, child accounts must be added or removed, usage and progress must be tracked, age-appropriate content must be ensured, and the platform must function properly.	Login is required (Admin credentials, full system access)
Members	Children who use the app to learn music and are between the ages of five and ten	A straightforward, vibrant, and user-friendly interface is necessary, as are audio-based prereader instructions, entertaining activities, immediate feedback, and badges to maintain motivation.	Logging in is necessary (simplified login, such as avatar + PIN)

6. Project Scope

Scope Area	Details
Main Purpose of the Web Application	To create a music class for children between the ages of 6 and 7, where the course is taught through an online interface by means of games,practice, and instrumental exploration.
Content to be Provided	Music lessons, ear-training exercises, instruction to pre-readers, virtual instrument interface
Learning Resources / Activities Included	• Virtual instrument • Note and melody matching activities • “Guess the sound” listening game
Features Available to Visitors	The about page, preview activity, general information for parents
Features Available to Registered Members	Simple login page, access to full lesson, interactive games, virtual instrument, progress tracking
Features Available to Administrators	Manage/add/remove member's account, upload and organize lessons and songs, monitor individual progress, review and approve content
Out of Scope / Features Not Included	Live tutoring, social features, advance music theory, mobile app version

7. Proposed Functionalities and Responsibility Allocation

Use this section to list all major functions of the system. Each function must have a person responsible for documentation, coding, and presentation.

No.	Functionality / Module	Description	User Type	Database Operation Involved	Person in Charge of Documentation	Person in Charge of Coding	Person in Charge of Presentation
1.	User login and registration	can create an account and securely log in to the website.	Normal User	ADD	CHANG JIR SHEN	CHANG JIR SHEN	CHANG JIR SHEN
2.	User Profile Management	can view and update their profile information.	Admin and Normal User	SELECT, UPDATE	MISHA A/P PATHMARAJAH	MISHA A/P PATHMARAJAH	MISHA A/P PATHMARAJAH
3.	Subscription Management	can view subscription plans, subscribe to a plan, and access subscription-based content.	Normal User	SELECT	RUVIEKA A/P M BALAKRISHNAN	RUVIEKA A/P M BALAKRISHNAN	RUVIEKA A/P M BALAKRISHNAN
4.	Music Theory Learning Content	can access interactive learning materials covering basic music theory.	Normal User	SELECT	GAN ZHI MING	GAN ZHI MING	GAN ZHI MING

5.	Interactive Music Game	can play educational games designed to practise music theory.	Normal User	SELECT	CHANG JIR SHEN	CHANG JIR SHEN	CHANG JIR SHEN
6.	Difficulty Selection	can choose different game difficulty levels such as Easy, Medium, and Hard.	Normal User	SELECT	GAN ZHI MING	GAN ZHI MING	GAN ZHI MING
7.	Game Score and Progress Tracking	records game scores and learning progress for each user.	Normal User	SELECT	BHAVANPREET KAUR SOHI A/P JASBIR SINGH	BHAVANPREET KAUR SOHI A/P JASBIR SINGH	BHAVANPREET KAUR SOHI A/P JASBIR SINGH
8.	Admin Dashboard	can securely access the administration section of the system.	Admin	SELECT	MISHA A/P PATHMARAJAH	MISHA A/P PATHMARAJAH	MISHA A/P PATHMARAJAH
9.	Game and Learning Content Management	can add, edit, and delete music theory learning materials , educational games and game questions.	Admin	SELECT, UPDATE, DELETE	RUVIEKA A/P M BALAKRISHNAN	RUVIEKA A/P M BALAKRISHNAN	RUVIEKA A/P M BALAKRISHNAN
10.	Progress Monitor	can view users' game scores and learning progress.	Admin	SELECT	CHANG JIR SHEN	CHANG JIR SHEN	CHANG JIR SHEN
11.	Subscription Management	can create and manage subscription plans and	Admin	SELECT, UPDATE, DELETE	MISHA A/P PATHMARAJAH	MISHA A/P PATHMARAJAH	MISHA A/P PATHMARAJAH

		monitor subscription status.					
12.	System Navigation and Interface	Provide simple navigation and an attractive, easy-to-use interface suitable for children and administrators.	Admin and Normal User	n/a	BHAVANPREET KAUR SOHI A/P JASBIR SINGH	BHAVANPREET KAUR SOHI A/P JASBIR SINGH	BHAVANPREET KAUR SOHI A/P JASBIR SINGH

8. Proposed Database Tables

No.	Table Name	Purpose	Main Data Stored
1	Users	Stores data of the accounts for children as well as superadmins.	User ID, username, email, password(hashed), role, registration date, account status
2	LearningContent	Stores data of music theory learning contents for the children.	Content ID, Title, Description, content type, difficulty level, created by (admin), status
3	Games	Stores data of the games based on the music theory.	Game ID, game name, description, difficulty level (easy/medium/hard), status
4	GamesQuiz	Stores data of the specific question/round of each game.	Question ID, Game ID, question text, answer options, correct answer
5	GameProgress	Stores data of the scores and learning progress of each user in games.	Progress ID, User ID, Game ID, attempts, date of completion, score
6	SubscriptionPlans	Stores data of the plans which can be subscribed by the users.	Plan ID, plan name, price, duration (monthly/yearly), description
7	UserSubscriptions	Stores data of the plan of each user and the status of that plan.	Subscription ID, User ID, Plan ID, start date, end date, status

9. Proposed Project Schedule

Week Date	Planned Task	Person Responsible
Week 3, 17–23 August 2026	Receive and review the group assignment requirements and discuss possible project ideas	All group members
Week 4, 24–30 August 2026	Select the Muzik 67 project idea and determine its objectives, mission, target audience and scope	All group members, led by Ruvieka A/P M Balakrishnan
Week 5, 31 August–6 September 2026	Confirm group membership and allocate the project modules and responsibilities	All group members
Week 6, 7–13 September 2026	Complete group registration by 8 September and begin preparing the project proposal	All group members, led by Ruvieka A/P M Balakrishnan
Week 7, 14–20 September 2026	Prepare the functionalities, database tables, workload distribution and proposed project schedule	All group members
21–27 September 2026	Review, combine and improve all sections of the proposal report	All group members
Week 8, 28 September–4 October 2026	Finalise and submit the project proposal by 4 October 2026	All group members, led by Ruvieka A/P M Balakrishnan
Week 9, 5–11 October 2026	Create the database structure, basic website layout, navigation and user interface	Bhavanpreet Kaur Sohi and all group members
Week 10, 12–18 October 2026	Develop user registration, login, profile management and the admin dashboard	Chang Jir Shen and Misha A/P Pathmarajah
Week 11, 19–25 October 2026	Develop subscription management, music theory content and difficulty-selection features	Ruvieka A/P M Balakrishnan and Gan Zhi Ming
Week 12, 26 October–1 November 2026	Develop the interactive music game, score tracking, progress tracking and content-management features	Chang Jir Shen, Bhavanpreet Kaur Sohi and Ruvieka A/P M Balakrishnan

Week 13, 2–8 November 2026	Integrate and test all modules, fix errors, complete the final report and submit the final report and web project files by 8 November 2026	All group members
Week 14, 9–15 November 2026	Present and demonstrate the completed Muzik 67 web application	All group members

10. Summary of Workload Distribution

Student Name	**No. of Main Modules**	**Main Module Areas**	**Workload Percentage**
BHAVANPREET KAUR SOHI A/P JASBIR SINGH	2	Razor Page, C# Coding, CSS styling, Database Operation	20%
CHANG JIR SHEN	3	Razor Page, C# Coding, CSS styling, Database Operation	20%
GAN ZHI MING	2	Razor Page, C# Coding, CSS styling, Database Operation	20%
MISHA A/P PATHMARAJAH	3	Razor Page, C# Coding, CSS styling, Database Operation	20%
RUVIEKA A/P M BALAKRISHNAN	2	Razor Page, C# Coding, CSS styling, Database Operation	20%
Total			100%